

EL YOUSFI HAJAR

TELECOMS, EMBEDDED SYSTEMS & AI ENGINEERING STUDENT (M2)

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Availability : 6-month Final-Year Internship (PFE) starting February 2027 — Casablanca / Rabat

PROFESSIONAL SUMMARY

Master's student in ISERT engineering (FSTM), dual profile in AI / embedded systems & telecoms, results-oriented: real deployed solutions (YOLOv8 detection at 91–93%, real-time Raspberry Pi camera, O-RAN 5G xApp, OCP industrial monitoring). Rigorous, autonomous and comfortable working from sensor to network, seeking a 6-month Final-Year Internship (PFE, February 2027) to deliver concrete value in AI, embedded systems and networks.

EDUCATION

FSTM – Master's in Embedded Systems, Networks and Telecommunications Engineering (ISERT) 2025 – 2027

- 2nd year (M2): embedded systems, networks & telecoms, image processing & vision, artificial intelligence.

FSTM – Bachelor's in Computer Science and Multimedia Networks (IRM) 2024 – 2025

FSTM – DEUST in Computer Science 2022 – 2024

Ain Harrouda High School – Baccalaureate in Physical Sciences (French Option) 2021 – 2022

INTERNSHIP

Proactive Load Balancing preserving QoE using Deep Learning (O-RAN 5G) | FSTM Jun – Aug 2026

- Predictive load-balancing xApp for the O-RAN Near-RT RIC: LSTM/GRU prediction (92% accuracy) + DRL agent (PPO), with handover decisions through the E2 interface (E2SM).

Design of a Smart Intrusion Detection Camera (Raspberry Pi, Embedded Deep Learning) | FSTM Jun – Aug 2026

- Real-time people detection on Raspberry Pi 4 (PFA): YOLOv8n, 3-level alerts, GPIO (LED/buzzer) and MQTT publishing.

MENPS – IT Directorate (Bachelor Internship) | Rabat Apr – Jun 2025

- Development of the Mostaqbali application (React.js, Spring Boot) for the Ministry of National Education: application management, tracking, validation, securing data flows and automating the administrative process.

PROJECTS

AI Safe Drive Guardian – Morocco-Localized Driving ADAS | Python, OpenCV 4, Dlib 19, YOLOv8n, Twilio 2026

- Drowsiness (EAR 91%), fatigue/yawning, phone use (93% at 37 FPS CPU) and distraction; audio alerts in Darija and emergency SMS (Twilio).

Smart Monitoring STEP OCP Khouribga – anomaly detection | Python, NumPy, Pandas 2026

- Isolation Forest + business thresholds on 171 measurements of an OCP water treatment station (volume, pH, turbidity): 60 anomalies flagged.

WebSCADA Pro – Intelligent Industrial Monitoring | React.js, Node.js, Socket.io, InfluxDB 2026

- Real-time web SCADA: dashboard, animated synoptic (SVG), 3-level alarms, 24h trends, OEE calculation; WebSocket streaming (latency < 15 ms).

TECHNICAL SKILLS

- **Telecoms & networks** : 2G to 5G, O-RAN (Near-RT RIC, xApp, E2), CDMA, LTE, FTTH
- **AI & data** : Python, PyTorch, scikit-learn, LSTM/GRU, PPO, Isolation Forest
- **Embedded systems** : C, C++, Raspberry Pi, GPIO, sensors, PROTEUS
- **IoT & real-time** : MQTT, WebSocket, FastAPI, SCADA
- **Embedded vision & deep learning** : OpenCV, YOLOv8, Dlib, CNN
- **Development** : React.js, Spring Boot, Java, SQL, Git, UML, MATLAB, InfluxDB

CERTIFICATIONS

- CCNA Network Security – Cisco
- Network Defense – Cisco
- Claude AI – Anthropic

LANGUAGES & ACTIVITIES

- **French** : Fluent, **English** : Fluent, **Arabic** : Native, **Italian** : Beginner, **Spanish** : Beginner, **Chinese** : Beginner
- **Activities** : Open Data Club (open data and analysis projects, FSTM 2026); Robotings Club (robotics and embedded electronics, workshops and tech competitions, FSTM 2023 – Present).